Causal Analysis of Graph Signals for Brain Effectome Inference

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***Abstract*—Understanding the directed interactions between brain regions is critical for analyzing neuro-degenerative diseases like Alzheimer's (AD). Traditional functional connectivity (FC) methods capture statistical associations but fail to infer causal relationships, limiting their ability to reveal structural disruptions in disease progression. In this exploratory work, we propose a causal graph inference and spectral analysis framework for EEG signals. Specifically, we leverage Granger causality and spectral graph methods to construct and analyze the effective connectome (effectome) of the brain. Our work reveals that AD networks exhibit lower algebraic connectivity (λ_2), reduced modularity (eigenvalue gaps), and increased structural sensitivity to perturbations, compared to healthy individuals. Additionally, a study of temporal dynamics over the inferred networks shows reduced and local node volatility patterns over time for AD brain networks.**

I. Introduction

Network based representations of the brain ("connectome") are crucial for modeling brain function, cognition, and disease. In a connectome model, the brain regions are treated as nodes, and their structural or functional connections as edges, yielding a graph of the organization of brain network [1].

While structural connectivity is often inferred through brain imaging, functional connectivity (FC) is determined using the statistical associations between activity time-series from different brain regions. FC has become a widely used measure to characterize brain networks at rest and during tasks [2]–[5]. However a well recognized limitation is that FC, by its nature, is undirected and non-causal. This means that while it can capture the coherence between two signals (i.e., whether they fluctuate together), it tells us nothing about whether activity in region A drives that in region B or vice versa [6]. In other words, it does not reveal the direction of information flow or underlying causal influences, motivating the use of advanced methods to infer directed interactions.

Effective connectivity (EC) aims to overcome this limitation by modeling the directed influence one neural system exerts on another [7], [8]. Establishing this cause-effect relationship between brain regions is particularly important when studying second-order effects of local degeneration, such as in Alzheimer's Disease (AD), where dysfunction originates in specific region but affects global network behavior [9]. Interpreting disease as a disruption in network-wide causal influence, rather than localized failure, offers a promising framework for studying disconnection syndromes [10]. EC further enables causal interventions to probe how global brain behavior might change if specific connections are disrupted, a scenario that closely mirrors the pathophysiological impact of AD and other neuro-degenerative conditions.

Our work builds on prior efforts [11]–[16] by developing a modular framework that estimates multivariate causal connectivity from EEG and extracts interpretable features from the resulting directed graphs to characterize disease-related network disruptions. We define the "effectome" here as the inferred observational effective connectivity, distinct from interventional definitions [17].

Our investigation is structured around the following central research questions.

- **RQ1**: Can we reliably generate a connectivity model of the brain by applying causal methods on EEG time-series data?
- **RQ2**: Can we use spectral graph analysis to identify the differences in properties of the brain network within healthy individuals and patients with a neuro-degenerative disease (specifically, Alzheimer's Disease)?

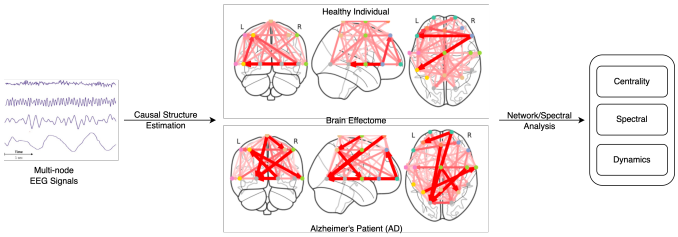


Fig. 1. System Architecture. We estimate the causal structure of the brain from EEG signals, and run analyses to study its properties.

These questions remain broad, and are intended as a framing device for our work. While the methods are designed to be generalizable, the current work only discusses them in the light of one dataset, thereby constraining the graph size to the number of regions within that data.

To address these questions, we present a modular pipeline, as shown in Figure 1. EEG data are first preprocessed and filtered, followed by multivariate Granger causality to infer directed influence networks (addressing **RQ1**). These graphs are analyzed using tools from spectral graph theory (including Hermitian Laplacians and Jordan decompositions), then probed using simulated diffusion and volatility analysis to

measure sensitivity to perturbation (for **RQ2**). Our approach is not only interpretable and grounded in signal-level causality, but is also designed for reuse and adaptation, making it suitable for broader applications in clinical and cognitive neuroscience.

Some of the key findings are given below:

- Despite individual variability, we observe distinguishable patterns in the effectome between healthy and AD groups, particularly in connectivity strength and directionality
- Spectral analysis (specifically the Fiedler value, λ_2) suggests that healthy networks are more densely connected and robust to perturbations (changes in structure), while the AD networks are more fragmented and sensitive to structural change
- Temporal dynamics indicate that AD networks exhibit reduced node/edge volatility and greater signal stationarity, both signatures consistent with diminished network adaptability

II. Data, Experiments & Results

For this work, we consider the EEG dataset from Vichietti (MV) [18]. The presented results are from tests for a randomly chosen subject in both the healthy ($n_H = 12$) and AD groups ($n_{AD} = 80$). Specifically, the data has 19 nodes, with 1024 samples acquired at resting state (eyes closed) with a sampling rate of 128Hz, and total duration 8 seconds. We acknowledge that scalp-level Granger Causality may be confounded by volume conduction [19], and that short recording durations limit the estimation of low-frequency dynamics [20]. Consequently, the connectivity patterns presented here should be interpreted as illustrative of the spectral framework's capabilities rather than definitive clinical characterization of cortical pathology.

A. Preliminaries

Consider $x_n[t]$, a discrete time series on node v_n in graph $G = (V, E, W)$, where n indexes the nodes of the graph and t the time samples. Let N be the total number of nodes and K be the total number of time samples, and $x[t] \in \mathbb{C}^N$ represents the graph signal at time t. We generate the effectome graph G, with edges E and weights W determined through statistical tests. Within W, w_{ij} represents the weight of the edge between nodes i and j.

B. Network Construction

We construct the effectome weight matrix W through causal inference. While multiple methods exist, a popular method is Granger Causality (GC) [21], [22]. We check if a variable X *granger causes* another variable Y, by determining if the inclusion of past observations (determined by maximum lag) of the cause reduces the prediction error of the effect using (multi-variate) auto regressive (MVAR) models [23]–[25]. At the most basic version, we pick the elements pairwise, and test the strongest direction of the causal effect [26].

We notate the causation as $X \xrightarrow{G} Y$. The null hypothesis H_0 remains that one variable does not granger cause the other.

To run the tests, we build two models, without (as shown in equation 1) and with (equation 2) X_t, the variable X at time t.

$$Y_t = \alpha + \sum_{i=1}^{L} \beta_i Y_{t-i} + \epsilon_t \tag{1}$$

$$Y_t = \alpha + \sum_{i=1}^{L} \beta_i Y_{t-i} + \sum_{i=1}^{p} \gamma_i X_{t-i} + \epsilon_t \tag{2}$$

Here, L is the max-lag order, α is the intercept term, β_i & γ_i are the coefficients of the lagged values of Y_t and X_t respectively, and ϵ_t is the error term.

We compare these models using hypothesis tests like the F-test or the Wald method and reject outcomes with $p \geq 0.05$. These equations can be extended to include other control variables Z_t, as is typical in real-world networks.

For the MV dataset, given the sampling frequency, we pick a max lag of 10 in line with prior works [27], [28].

The heatmaps are presented in Figure 2. We observe a lack of connectivity from the T6 and T5 regions (posterior temporal and parietal occipital junctions) in AD, which signifies impaired memory retrieval and language comprehension. F8 and Fz have higher connectivity, suggesting possible compensatory mechanisms at play. Generally, while the healthy network is more uniform, we observe stronger causation in AD brain networks, showing increased and disorganized connectivity likely due to pathological processes resulting from the disease (figure 1). These results are also in line with prior research in the field [29]–[31]. As a part of ongoing work, we are looking into other popular information-theoretic measures like Transfer Entropy (TE) [32], [33] and Permutation Entropy [34]. Preliminary results with TE (Figure 3) show sparser AD brain network compared to healthy brain networks.

C. Network Analysis

We analyze structural and dynamical properties of the inferred brain network G using centrality metrics, spectral decomposition [35]–[37], and temporal volatility. Specifically, we are interested in studying the Fiedler value (λ_2).

1) Centrality measures : Figure 4 shows the centrality measures grouped by brain regions. Typical brain activity happens in the posterior regions, but a higher eigenvector centrality in case of AD patients in the frontal region suggests a compensatory mechanism, likely due to loss of function in the posterior regions [29]–[31].

2) Spectral Analysis: Spectral properties of the Hermitian Laplacian [38], [39] of G reveals key structural trends, especially when perturbed. We use the Hermitian to preserve directionality.

Healthy brain networks show higher low-index eigenvalues and greater eigengaps, suggesting stronger modularity and robustness to perturbation. This characteristic is quantified by the Fiedler value λ_2, with healthy subjects showing $\mu = 0.44, \sigma = 0.23$. On the other hand, for the AD brain network, the eigenvalues are more closely spaced, suggesting

reduced connectivity or weakened modularity in the network ($\lambda_2 : \mu = 0.38, \sigma = 0.13$). This may reflect the loss of functional integration across brain regions. With Transfer Entropy, the λ_2 values are 0.83 and 0.57 for healthy and AD brains respectively, which remain largely consistent with our prior observations. These patterns suggest reduced global connectivity in AD, but with high within-group variability.

Notably, bootstrap-based group comparisons with over 10,000 stratified samples revealed low statistical significance: only 0.03% of t-tests and 2.1% of U-tests were significant at the edge level. This lack of group-level convergence likely reflects the high heterogeneity of scalp EEG and the limitations of short-duration recordings discussed above. However, our analysis highlights that edge-level directional differences and localized spectral features (like reduced eigengaps, lower node volatility, frontal compensatory hubs) still capture meaningful network disruption in Alzheimer's disease. These findings motivate a shift toward finer-resolution, region specific, and dynamic models of brain connectivity in neuro-degeneration.

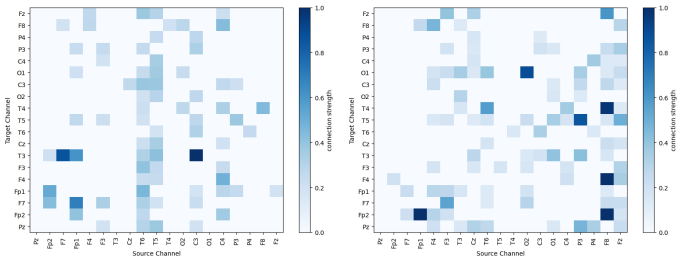
3) Temporal Dynamics: Multi-layer networks constructed using a rolling-window analysis (window size = 256, step size = 128) enable us to examine edge and node volatility over time. To further explore temporal variability in brain activity, we analyze and visualize the results topographically (Figure 5). Volatility was quantified as the variance of node activity across samples, reflecting the dynamic reconfiguration capacity of each brain region.

Complementing this, Figure 6 shows the temporal evolution of volatility across successive windows. Healthy brains demonstrate rich spatial transitions in volatility over time, indicative of dynamic state changes, whereas AD networks exhibit restricted, temporally flat patterns. Together, these results support the hypothesis that AD disrupts the network's ability to adapt and reconfigure, leading to more rigid and less responsive functional dynamics.

In addition, our analysis of stationarity revealed that healthy EEG recordings were non-stationary in 11 out of 19 channels, with several nodes displaying significant temporal fluctuations, whereas AD signals were predominantly stationary across nodes. The presence of non-stationarity in healthy brains points to richer temporal dynamics and greater functional adaptability, whereas the stationary nature of AD recordings suggests a pathological stabilization and diminished capacity for network reconfiguration [40]. While volatility quantifies spatial variability in node activity, stationarity assesses temporal autocorrelation. Together, these results support the hypothesis of reduced dynamical flexibility in AD.

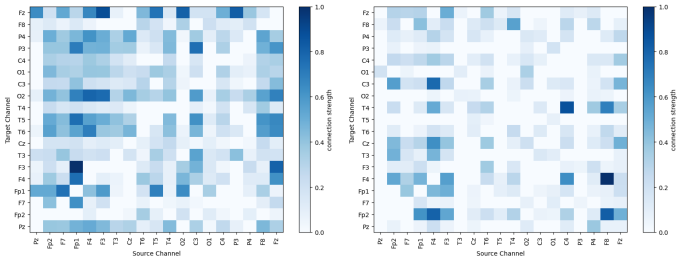
III. Conclusion & Future Work

This work presents a causal, graph-theoretic approach to modeling effective connectivity in brain networks derived from EEG time-series data, comparing healthy individuals and those with Alzheimer's Disease (AD). We infer directed brain networks using Granger Causality and spectral graph theoretic methods, and examine their topological and temporal characteristics.



(a) Healthy Individual (b) AD Patient

Fig. 2. Heat map of effective connectome strength, obtained using Granger Causality. We can observe a lack of connectivity from the T6 and T5 regions (posterior temporal and parietal occipital junctions) in AD, which signifies impaired memory retrieval and language comprehension. F8 and Fz have higher connectivity, which shows possible compensatory mechanism at play



(a) Healthy Individual (b) AD Patient

Fig. 3. Heat map of effective connectome strength, obtained using Transfer Entropy. Each heatmap shows directed information transfer between EEG channels, with sources along the x-axis and targets along the y-axis. AD brains exhibit sparser and weaker transfer entropy connections compared to healthy brains, suggesting disrupted directed functional connectivity.

Our results remain in-line with prior work, and indicate that AD brain networks exhibit reduced algebraic connectivity, distinguished modular structure, and lower temporal volatility, consistent with weakened integration and fragility in dynamical organization. However, these trends are not uniformly significant at the group level, reflecting both intra-group heterogeneity and the challenges of reliable graph inference from noisy neural signals. Despite this, edge-level differences and spectral signatures reveal localized disruptions that remain consistent across multiple analyses.

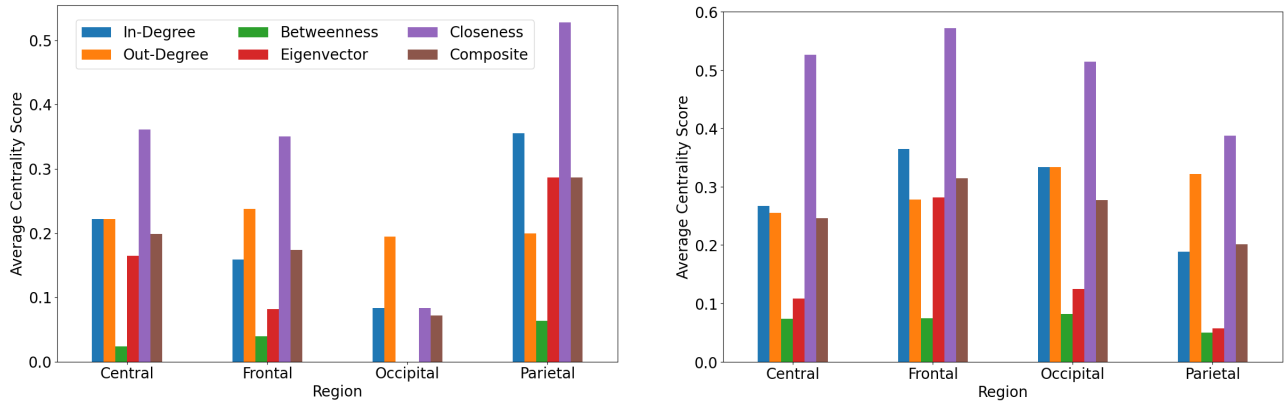
From a network-theoretic standpoint, this work emphasizes:

- Interpretable spectral metrics (e.g., λ_2) as a proxy for diffusion speed and connectivity resilience
- Multiscale volatility as a dynamic network feature
- The use of multilayer graphs to model time-evolving topologies

The modularity of our pipeline permits the replacement or extension of each stage (e.g., network inference, spectral analysis, temporal dynamics), enabling broad generalization to other neuroimaging modalities and networked systems.

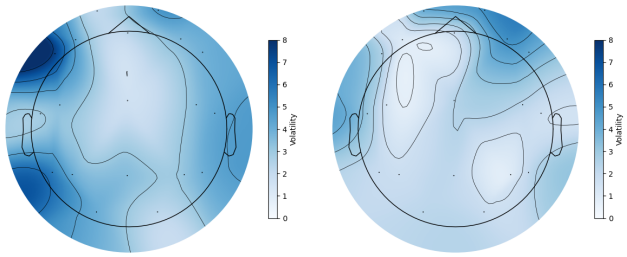
To this end, moving forward, we aim to:

- Extend the analysis to BOLD-fMRI and task-based EEG, enabling multi-modal, multi-timescale inference



(a) Healthy Individual (b) AD Patient

Fig. 4. Centrality measures grouped by regions. Typical brain activity happens in the posterior regions, but a higher eigenvector centrality in case of AD patients in the frontal region shows a compensatory mechanism at play, likely due to loss of function in the posterior regions



(a) Healthy Individual (b) AD Patient

Fig. 5. Topographical maps of node volatility across the scalp for (a) healthy individual and (b) AD patient. Healthy individuals exhibit broader and higher volatility levels, while AD patients have it reduced and more spatially localized, indicating a disruption in dynamic network stability.

- Incorporate graph learning techniques (e.g., structured sparsity, GNN-based priors) to regularize noisy causal graphs and extract interpretable motifs [41]
- Improve the robustness of causal inference by addressing non-stationarity and validating results using information-theoretic measures [42]
- Employ causal perturbation strategies to study graph responses, enabling a principled way to assess fragility beyond static metrics
- Model cross-frequency coupling and dynamic reconfiguration via frequency-layered multilayer networks, linking network geometry with signal content [40]
- Fit the resulting temporal graphs to low-dimensional latent models to quantify transitions and capture underlying dynamical regimes

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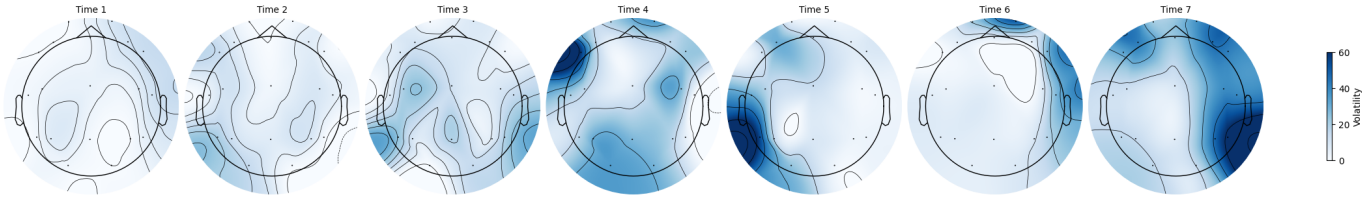
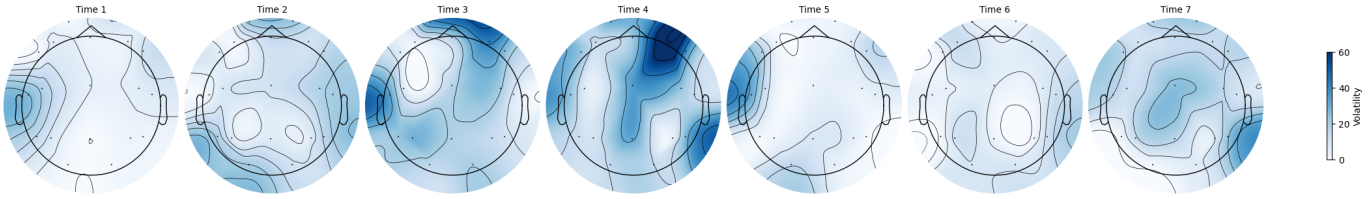


(a) Healthy Individual



(b) AD Patient

Fig. 6. Temporal evolution of node volatility across sliding windows for (a) Healthy individual and (b) AD patient. Each scalp map represents node-wise volatility at a specific time window. Healthy individuals display more spatially diverse and dynamic changes in volatility over time, while AD patients exhibit more local and reduced volatility patterns.

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